

Approximate Bayesian Detection of Colibactin Driven Mutations in Colorectal Cancer

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This is an edited extract from a larger group project. All of the work here is my own.

1. Introduction. Exposure to colibactin, a genotoxin produced by the *pks+* *Escherichia coli* (E. coli) bacterium, has recently been proposed as a contributing factor to the rising incidence of early-onset colorectal cancer [5]. This report extract summarises the methods used to develop a novel classification framework to detect colibactin exposure in mutational signatures.

2. Biological assumptions. Cancer tumours are understood as being driven by the accumulation of somatic mutations. These mutations may arise from a variety of sources; consider, for example, the well known hazards of tobacco smoking and UV exposure, or natural clock-like processes that accumulate with age. Each process tends to leave a characteristic pattern of mutations across the genome known as a mutational signature, with colibactin induced DNA damage associated with signatures SBS88 and ID18 [8].

Our focus lies on the adenomatous polyposis coli (APC) gene, a tumour suppressor that is the most frequently mutated gene in colorectal cancer and whose loss is typically an early event in tumour development. Indeed, mutations associated with colibactin have been found to be enriched in the splice-site variant c.835-8A>G [8, 5, 6], and thus SBS88 is of more interest than ID18.

Single base substitutions, when classified by their trinucleotide context, can be seen as a 96-dimensional vector. After normalisation this becomes a probability distribution over these 96 contexts; refer to Figure 1 or the larger COSMIC database [10]. For modelling purposes, the mutational profiles of the epithelial cells lining the colon (the primary site associated with colorectal cancer) are assumed to be a linear mixture of healthy background mutations and a colibactin-driven component. Following the literature on normal colonic crypts [7], a normal profile is composed of signatures SBS1, SBS5, and SBS18, weighted

$$P_{\text{normal}} = 0.4\text{SBS1} + 0.4\text{SBS5} + 0.2\text{SBS18}. \quad (1)$$

Both SBS1 and SBS5 are clock-like signatures, whereas SBS18 indicates oxidative DNA damage [2, 3]. The profile of an individual with colibactin-induced damage is controlled through the injection level parameter α ,

$$P_{\text{exposed}}(\alpha) = \alpha\text{SBS88} + (1 - \alpha)P_{\text{normal}}. \quad (2)$$

Publicly available colorectal cancer data is limited, so a large number of synthetic mutational profiles are generated by sampling from P_{exposed} . An important piece for this

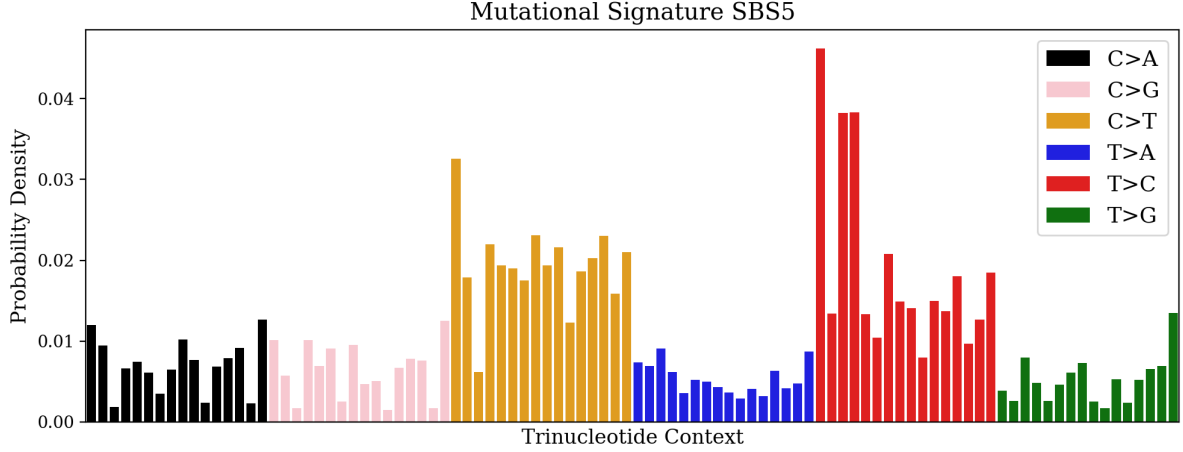


Figure 1: The characteristic pattern of mutations that describe the common clock-like signature SBS5. The context labels, e.g. A[C>A]A, are omitted for visual clarity, with mutations instead coloured by base substitution.

sampling is the injection level α . An analysis of colorectal tumours from the TCGA dataset [11] with non-negative least squares attribution (see e.g. [4, 9]) suggests that $\alpha = 0.0841$ is a reasonable estimate.

3. Classifier. With the ability to generate a large amount of synthetic data, inference on the injection level α is performed using principles from approximate Bayesian computing. Given an observed mutational profile, the framework proceeds as follows.

1. Define a discrete, uninformative (uniform) prior for α on $[0, 1]$.
2. For each value of α :
 - (a) Generate a large number of synthetic mutational profiles according to P_{exposed} .
 - (b) Compare each synthetic profile to the observed profile using a similarity metric. If the two profiles are sufficiently similar, accept α as a candidate value. Otherwise reject this value of α .
3. Normalise the number of acceptances across all values of α to form a posterior distribution for α .
4. Use the posterior distributions to determine if there is significant evidence of SBS88 in the observed mutational profile.

There are a number of choices here to discuss. The similarity metric is chosen to be the cosine distance, a standard choice in mutational signature analysis [1, 10]. Its main advantage over alternatives, e.g. the Kullback-Leibler divergence, is its robustness to zero entries which can occur frequently in mutational profiles with few mutations. Moreover, at least $B = 1000$ batches of synthetic profiles are generated, with acceptance scores averaged over multiple trials. For confidence in the posterior, a threshold is set on the minimum number of acceptances needed for a given value of α to be considered.

Once the posterior is formed, its 90% highest density interval $[\alpha_0, \alpha_1]$ is computed and used to classify the presence of SBS88. The following proposal is a conservative rule that has empirically performed well.

1. If $\alpha_0 = 0$ then there is insufficient evidence that SBS88 is present.

2. If $\alpha_0 > 0.05$ there is sufficient evidence that SBS88 is present.
3. If $0 < \alpha_0 \leq 0.05$, then the upper endpoint α_1 and posterior mode $\hat{\alpha}$ are considered.
 - (a) If $\alpha_1 > 0.1$ and $\hat{\alpha} > 0.05$, there is sufficient evidence for SBS88.
 - (b) Otherwise the presence of SBS88 is again rejected.

A key part of the classifier not yet discussed is the similarity tolerance: the threshold that determines when two profiles are ‘sufficiently similar’. It is certainly a sensitive choice in the framework, for when it is too strict not enough of the B batches are accepted and no posterior is produced. On the other hand, if it is too relaxed, too many values of α are accepted and the inference loses its accuracy.

To tune the tolerance, a wide range of candidate (α, M) pairs are set, where M indicates the total number of mutations in a given profile. Then, for each pair, the classifier is run on a batch of synthetic profiles generated according to $P_{\text{exposed}}(\alpha)$, and again on a batch of synthetic profiles according to P_{normal} . Detection of SBS88 in the former case is recorded as a true positive, whereas detection in the latter case is recorded as a false positive. These are computed for a range of tolerance values $\text{tol} \in [0, 0.5]$, averaged over many trials. The optimal tolerance is chosen as that which maximises Youden’s J statistic,

$$\text{tol}_{\text{opt}}(\alpha, M) = \arg \max_{\text{tol}} [\text{TPR}(\text{tol}, \alpha, M) - \text{FPR}(\text{tol}, \alpha, M)].$$

The parameter space for the classifier can grow quickly, so care is taken to keep the implementation efficient. For instance, distance metrics are computed only once per (α, M) pair, computations are vectorised, and results are stored locally for later use.

4. Model validation. The classifier’s performance is assessed by applying it to synthetic profiles generated according to $P_{\text{exposed}}(\alpha)$. Its task is to then recover α .

Experiments suggest that it performs very well when there are at least $M = 100$ mutations, with α estimates being within 10% of the true value in all cases when $M \geq 100$ is known. Estimates on profiles with $M < 100$ were very noisy and unreliable, as expected. Importantly, it confidently denied the presence of SBS88 for every single healthy profile (profiles with $\alpha = 0$) irrespective of M .

5. Application to real data. Finally, the developed framework is applied to the mutational profiles of real colorectal cancer patients. This data comes from the cBioPortal data set and can be limited: for instance, the first cohort to consider are those patients who possess the specific c.835-8A>G mutation, of which there are only 35 with a total of 202 recorded mutations. Thus mutations are assumed to be independent and identically distributed, considered together as one individual cohort, for an average of six mutations per patient is far too small for reasonable inference.

Preliminary analysis of the data in this cohort suggests the modelling assumptions in (1) and (2) are appropriate. Under this premise, the classifier confidently indicates the presence of SBS88 with an average predicted value of $\alpha = 0.35$. This puts further faith in the classifier: patients with this particular c.835-8A>G variant were chosen because the mutation has been attributed to SBS88 in 64.3% of colibactin-positive cases [5]. With the statistical analysis elsewhere in this project (edited from this summary) this is consistent with a link between early-onset colorectal cancer and colibactin exposure.

The second cohort of interest is much larger, containing information for any individual with colorectal cancer. It has over 800 patients and a combined total of over 9000 mutations. Unfortunately, the preliminary analysis does not suggest the assumptions (1) are reasonable here - there seems to be less weight on the T>C mutations than thought. The current framework will therefore refuse to provide any estimates for SBS88 injection because it would contribute to an already overestimated part of the spectrum.

To this end, the SigProfilerAssignment tool [4] is used to extract plausible signatures that make up the data. It suggests that SBS1 and SBS5 are still present, but puts a large amount of weight on SBS10a, SBS10b, and SBS15, three signatures linked to defective DNA repair. Using a simplified approximation of the SigProfilerAssignment output, the combination

$$P_{\text{CRC}} = 0.3\text{SBS1} + 0.2\text{SBS5} + 0.1\text{SBS10a} + 0.15\text{SBS10b} + 0.25\text{SBS15} \quad (3)$$

is now taken as the assumed underlying distribution of mutations in colorectal cancer patients. The framework is otherwise the same with $P_{\text{exposed}}(\alpha) = \alpha\text{SBS88} + (1 - \alpha)P_{\text{CRC}}$. The similarity tolerance is retuned now that the underlying distribution has changed, and simple model validation is rerun before the classifier is run on the real data.

With a fair amount of work in merging different datasets (elsewhere in this project, removed from this summary), it is possible to split this large cohort into an early-onset cohort and otherwise. The results are interesting. Neither cohort suggests that SBS88 injection provides an improved fit compared to the baseline distribution (3), both providing a median estimate of $\alpha = 0$. However, estimates for the early-onset cohort are certainly higher than the other cohort. Refer to Figure 2, for example, where the mean, upper quartile, and outlier estimates are notably higher in the early-onset cohort than not. This is an interesting find but is by no means strong statistical evidence. The pronounced skew in distribution may be entirely attributable to the small cohort with the splice-site mutation c.835-8A>G, for example.

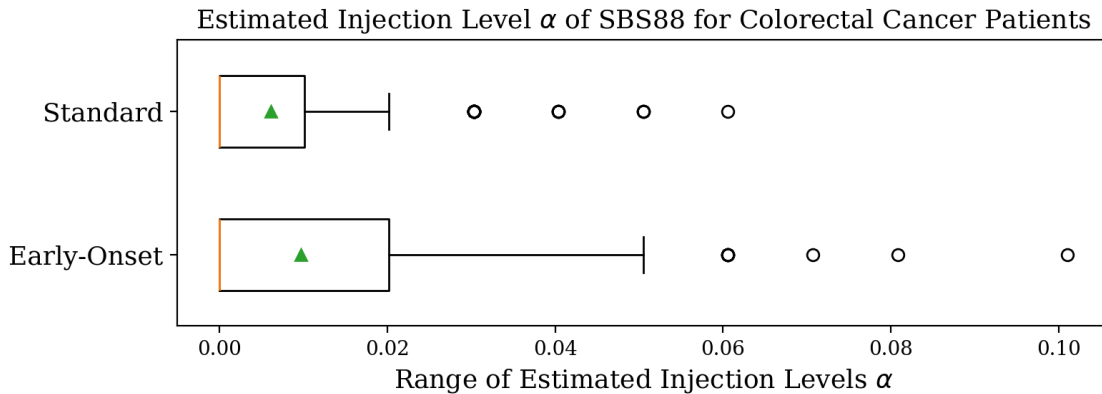


Figure 2: The estimated injection level α of mutational profiles from colorectal cancer patients. Estimates are computed from 500 subsamples of $M = 250$ mutations drawn, with replacement, from the pooled cohorts, assuming the underlying distribution in (3). The mean is indicated with the triangle, and upper whiskers are taken as $Q_3 + 1.5\text{IQR}$.

6. Concluding discussion. This extract has developed an approximate Bayesian computing based classifier for detecting SBS88 in mutational profiles. On synthetic data the

method performs well: it recovers the injection level reliably for $M \geq 100$ mutations and does not produce false positives on healthy profiles. Applied to real data, its conclusions are more modest. The small c.835-8A>G cohort shows clear evidence of SBS88, as is expected given the variant’s known attribution. In the large colorectal cohort the median estimate is $\alpha = 0$ for both early-onset and later-onset patients; the elevated upper tail among early-onset cases is suggestive but may be driven entirely by the splice-site subgroup.

One clear limitation of this work is the fixed baseline inference; inference on α is only as good as the assumed P_{CRC} , and the need to replace (1) with (3) well demonstrates this importance. A framework that treats the baseline as unknown rather than fixed is one alternative approach that could be instructive. Furthermore, it would be insightful to perform some analysis that attempts to attribute the difference in early and late-onset cohorts to the smaller splice-site mutation cohort, and how the removal of this smaller cohort influences the results.

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